Sell Spot

Submitted To:

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Course Name

Data Structure and Algorithem

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Data Structure and Algorithm Lab Manual (Week-11)

Lab Instructor: Ms. Rabeeya Saleem Session: 2024

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DSA PROJECT PROPOSAL

Sell Spot:

Sell Spot – A Classified Marketplace Platform

It is all ways a problem for peoples who want to sell their used goods, So Sell Spot is a Platform that facilitates the user to sell those things from home:

- Fake or misleading ads
- Slow searching and filtering
- Miscommunication between buyers and sellers
- Difficulty in organizing a large number of listings

This project develops a classified marketplace system named Sell Spot, that efficiently manages product listings, user data, and communication using Data Structures and Algorithms

Core Objective

- Use structured data to store ads and users
- Automate product listing and management
- Ensure no duplicate or fake ads
- Maintain clean categories for easy browsing
- Provide fast product search and filtering
- Organize user and product records properly

Scope Definition

Data structure need to be implemented:

Data Structure	Purpose in Project
Array	Stores product lists in categories (e.g., mobiles, cars, electronics) for quick access.

Queue	Manages user support requests or message notifications in first-come-first-serve order.
Linked List	Maintains dynamic lists of ads where users can add, update, or remove listings easily.
Stack	Provides undo functionality for actions like editing an ad or deleting a message.
Hash Table	Enables fast searching of products and users through hashed keywords or user IDs.
Graph	Represents connections between buyers and sellers, product recommendation paths.
Tree	Organizes categories and subcategories in a hierarchy.
Heap	Manages priority listings, ranking ads based on payment, views, or relevance.

Data Storage Platform Textfile or MYSQL

Detailed Feature Set

User Management

- Register new users
- Login & authentication
- Edit user profile
- Manage posted ads list
- Block/verify users (Admin)

Ad Management

- Create new product ads
- Edit existing ads
- Delete ads
- Category-wise ad storage
- Prevent duplicate ads

Product Catalog

- Add product categories
- Organize subcategories
- Category browsing
- Remove or update categories

Search & Filtering

- Keyword search (Hash-based)
- Category filter
- Price range filter
- Location-based filter
- Sort by relevance/date

Messaging System

- Buyer–seller messaging
- Message queue handling
- New message notifications
- Delete chat history

Admin Panel

- Approve/reject ads
- Remove fake or spam ads
- Manage user accounts
- Category and subcategory control

Technology Stack

Language	C#
Data Storage	MYSQL
UI	Winform